

KOMAL UPADHYAY

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Summary

Software Development Engineer skilled in backend microservices architecture, data engineering, and SQL optimization, with experience at Mercedes-Benz R&D and BNY Mellon. Specializes in enhancing event processing efficiency and system reliability while reducing deployment times and strengthening audit readiness. Eager to leverage technical skills to create significant value and advance business goals.

EDUCATION

Northeastern University

Master's, Computer Science

2023 - 2025

- **GPA:** 4.0
- **Coursework:** Data Structures, DBMS, Distributed Systems, Cloud Computing, Android Dev, Web Dev, AI/ML

Dr. A. P. J. Abdul Kalam Technical University

Bachelor's, Electrical Engineering

- **GPA:** 3.7
- **Coursework:** Embedded Systems, DBMS, Data Communication Networks

TECHNICAL SKILLS

- **Languages:** Java, Go, Python, JavaScript, HTML, CSS, Kotlin
- **Cloud:** AWS (S3, Lambda, Dynamodb, RedShift), Azure (DevOps, Data Lake, Key Vault, Event Hubs), GCP (Pub/Sub, FireStore)
- **Libraries and Tools:** Android SDK, React, Django, Bootstrap, Node.js, Netlify, Git Version Control
- **Databases:** MySQL, SQL Server, MongoDB, Firebase, PostgreSQL

PROFESSIONAL EXPERIENCE

Mercedes-Benz Research & Development North America

Software Development Engineer Backend - Internship

May 2024 - Dec 2024

USA

- Architected high-performance microservices in Go and gRPC, reducing system latency by 30% and tripling request throughput.
- Designed and deployed resilient data ingestion pipelines using Kafka, goroutines, and custom load-balancing strategies ensuring real-time delivery and unlocking a 150x increase in revenue-critical event processing
- Built production-grade observability stack with Datadog dashboards, Prometheus metrics, and actionable alerts, cutting incident detection and triage time by over 50%.
- Created and maintained API documentation using Swagger for service contract clarity and seamless collaboration.
- Automated full-stack multi-cloud provisioning across AWS and Azure using Terraform and Helm, shrinking environment setup time from hours to minutes. Led reliability and scaling initiatives across containerized deployments in Kubernetes.

Bank of New York Mellon

Lead Database Developer

Jun 2021 - Mar 2022

India

- Led a 5-member engineering team to roll out Liquibase-based CI/CD pipelines for SQL Server, integrating with Azure DevOps to automate database schema changes, reducing manual deployment errors and cutting production rollout time by 80%.
- Conducted security reviews and applied fixes for SQL injection vulnerabilities across 4 legacy financial applications, aligning database access patterns with modern secure coding practices and improving audit readiness.
- Reviewed and optimized over 500 slow-running SQL queries by analyzing execution plans and applying indexing strategies, improving data retrieval speeds by 20% and reducing strain on production workloads during peak hours.

Tata Consultancy Services

Software Development Engineer

Nov 2014 - May 2021

Japan /India

- Customized and automated workflows across 5 Infor CloudSuite ERP modules, streamlining approval chains and increasing data consistency by 30%, while reducing manual processing times by 40%.
- Identified and resolved SQL injection risks in 4 financial apps, improving system security posture by 70%.
- Executed SSIS data migrations with 99% accuracy and built SSRS/VB.NET dashboards, improving decisions by 25%

PROJECTS

Pawsome Pals: Social Media Android App

- Created a Java-based Android app with Firebase backend, Google Maps integration, and deep linking for seamless UX.
- Applied ML-based content ranking and used Firebase multithreading APIs for real-time data rendering.
- Handled debugging and crash reporting with Firebase Crashlytics to ensure production stability.

AutoPurgeX: Event-Driven Object Lifecycle Orchestrator on AWS

- Built an event-driven system on AWS using Lambda and S3 to enforce object quotas in real-time.
- Implemented SNS-SQS fanout and conditional IAM for fault-tolerant cross-service invocation.
- Automated infrastructure provisioning with Terraform and added CloudWatch alarms for observability.

SkillMatchPro: Django, React, GPT4, MySql

- Developed a scalable REST API with Django and a responsive React frontend, deployed on GCP with load balancing and auto-scaling.
- Integrated GPT-4 chatbot for natural language querying of MySQL, enhancing data interaction and accessibility.
- Documented endpoints with Swagger (YAML), and implemented CI pipelines with automated tests for stable delivery.

LogInsight360: Java, Spring Boot, SQL Server, React, Swagger, Splunk, Jenkins, Docker

- Built a full-stack internal tool for engineers to visualize and query real-time application logs aggregated from microservices.
- Developed RESTful APIs in Java Spring Boot, using Swagger (OpenAPI) and SQL Server for storing structured logs.
- Integrated with Splunk for log forwarding, anomaly detection, and real-time debugging insights; deployed via Jenkins CI/CD and containerized with Docker.

Certifications

- AWS Certified Data Engineer – Associate